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BLM20364E - BLM22332E HEURISTIC OPTIMIZATION ALGORITHMS
TERM PROJECT

MULTI-OBJECTIVE DIET OPTIMIZATION PROBLEM (MODP) EVOLUTIONARY
MULTI-OBJECTIVE OPTIMIZATION ALGORITHMS

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ABSTRACT

Dietary planning is a complex optimization problem that requires balancing multiple conflicting objectives simultaneously, including nutritional adequacy, user preferences, meal preparation time, and total cost. Constructing an optimal daily meal plan manually is difficult due to the large number of possible food combinations and nutritional constraints.

The proposed system recommends a personalized daily menu consisting of breakfast and lunch/dinner meals from a database containing 405 prepared food items. The optimization process considers user-specific nutritional requirements and food preferences while minimizing meal cost and preparation time. Four state-of-the-art Multi-Objective Evolutionary Algorithms (MOEAs), namely NSGA2, SPEA2, SMS-EMOA, and NSGA3, are implemented and compared.

The system is evaluated using two different user profiles: a non-vegetarian user and a vegetarian user. Nutritional constraints are defined according to Dietary Reference Intake (DRI) values, and infeasible solutions are penalized using a constraint-handling mechanism. Experimental results demonstrate the effectiveness of evolutionary optimization techniques for generating diverse and feasible diet plans while satisfying multiple objectives simultaneously.

Keywords: Multi-objective optimization, diet optimization, evolutionary algorithms, NSGA2, SPEA2, SMS-EMOA, NSGA3, nutritional planning, MOMKP.

1. INTRODUCTION

Dietary planning plays an important role in maintaining human health and preventing nutrition-related diseases. A balanced diet must satisfy daily nutritional requirements while also considering factors such as food preferences, meal diversity, preparation time, and economic affordability. However, manually creating an optimal meal plan is a highly complex task because numerous constraints and objectives must be satisfied simultaneously.

Traditional optimization approaches often focus on a single objective, such as minimizing cost or maximizing nutrient intake. In real-life scenarios, dietary planning involves multiple conflicting objectives. For example, meals with higher nutritional quality may increase cost or preparation time. Therefore, the diet planning problem can be naturally formulated as a Multi-Objective Optimization Problem (MOP).

In this project, the diet optimization problem is modeled as a Multi-Objective Multidimensional Knapsack Problem (MOMKP). The objective is to generate personalized daily menus composed of breakfast and lunch/dinner meals selected from a dataset of 405 food items. The optimization framework uses evolutionary algorithms to search for high-quality solutions under nutritional and dietary constraints.

Four well-known Multi-Objective Evolutionary Algorithms (MOEAs) are implemented and compared:

- NSGA-II
- SPEA2
- SMS-EMOA

The algorithms are tested using two different dietary profiles:

- User 1: Non-vegetarian
- User 2: Vegetarian

The results are analyzed based on solution quality, diversity, convergence, and constraint satisfaction.

2. PROBLEM DEFINITION

The objective of this project is to recommend an optimal daily menu that satisfies user nutritional requirements while simultaneously optimizing multiple criteria.

The optimization problem includes the following objectives:

1. Maximizing user food preference
2. Minimizing total meal cost
3. Minimizing meal preparation time

The problem is formulated as a Multi-Objective Multidimensional Knapsack Problem (MOMKP), where each food item contributes to several nutritional dimensions such as calories, protein, carbohydrates, fat, sodium, and fiber.

The optimization system must select appropriate food combinations while satisfying nutritional constraints and dietary restrictions.

Mathematically, the optimization problem can be represented as:

Minimize:

- $\text{Cost}(x)$
- $\text{PreparationTime}(x)$
- $-\text{Preference}(x)$

Subject to:

- Nutritional constraints
- Dietary restrictions
- Meal composition constraints

where x represents a candidate daily menu.

```
Minimize/Maximize: f1(x) =  $\sum x_i \cdot \text{preference}_i$            → MAX (mandatory)
                  f2(x) =  $\sum x_i \cdot \text{cost}_i$              → MIN (if chosen)
                  f3(x) =  $\sum x_i \cdot (\text{prepTime} + \text{cookTime})_i$  → MIN (if chosen)
                  f4(x) =  $\sum x_i \cdot \text{co2}_i$              → MIN (if chosen)

Subject to: RLL_j ≤  $\sum x_i \cdot \text{nutrient}(j,i)$  ≤ RUL_j   for j = 1..5
            x_i ∈ {0,1},    n = 405 food items
```

Table 1: Mathematical Model

3. DIVISION OF WORK

2121221022 - Muhammed Eren Koçkan - **db.py** - database connection / queries

2321021019 - Ahmet Emin Uğurlu - **chromosome.py** - decode function

2221221048 - Ahmet Mesut Yolcu - **fitness.py** - fitness function

2321021003 - Hakan Kocaman - **main.py** - algorithm execution / hypervolume analysis

Ç2025201009-Hazal Beşire Turhan - **visualize.py** - visualization / plots

4. DATASET

The dataset used in this project contains 405 prepared food items. Each food item includes several attributes required for optimization and nutritional analysis.

The dataset includes:

- Food name
- Meal category
- Nutritional values
- Cost information
- Preparation time
- User preference score
- foodGroups table explanation
- breakfast-compatible groups
- lunch/dinner-compatible groups
- duplicated groups explanation

The food items are divided into two major categories:

1. Breakfast meals
2. Lunch/Dinner meals

In the chromosome representation, the first 94 food items correspond to breakfast foods, while the remaining items correspond to lunch/dinner foods.

The dataset also contains nutritional information such as:

- Energy (kcal)
- Protein
- Carbohydrates
- Fat
- Fiber
- Sodium
- Vitamins and minerals

Table	Key Columns	Purpose
foods	id, name, foodGroupId, cost, preference, preparingTime, cookingTime, co2	Main food item list with objective values
user_foods	userId, foodId, preference	Per-user preference ratings (use this over foods.preference)
food_nutrients	foodId, nutrientId, value	Nutritional content per food item
nutrients	id, name, unit	Nutrient names and units
dri	userId, nutrientId, RLL, RUL	Daily intake lower/upper bounds — per user
food_group	id, name	18 food groups — used for diversity calculation

Table 2: MySQL database

Sol #	Pref ±	Cost ±	Time ±	Foods (top 5)	Nutrients vs DRIs
#1	-17.85	43.04	837	SEBZE CORBASI DONDURMA DONDURMA SALATLIK SODA BIRNAC FLYME SRANI ZELTIR	Energy(kcal): 2719 (2085.0-2400.0) # Protein(g): 45.86 (0-100.0) # Carb(g): 238 (175.0-399.0) # Fiber(g): 33.55 (0-9999.0) # SaturatedFat(g): 29.13 (1560.0-2900.0) # Groups: 12
#2	-41.11	24.09	175	SAGSIKA KORUNDUN TURSU DONDURMA SALATLIK SODA BIRNAC FLYME KURU KARTI	Energy(kcal): 2747 (2085.0-2400.0) # Protein(g): 107.88 (0-100.0) # Carb(g): 10 (175.0-399.0) # Fiber(g): 39 (0-9999.0) # SaturatedFat(g): 1.19 (1560.0-2900.0) # Groups: 8
#3	-39.23	24.33	119	MADAGILNA AYRINA CHOCOLATELUGAL (EVREK) PASTIRMA PASTIRMA	Energy(kcal): 2711 (2085.0-2400.0) # Protein(g): 107.88 (0-100.0) # Carb(g): 10 (175.0-399.0) # Fiber(g): 39 (0-9999.0) # SaturatedFat(g): 1.19 (1560.0-2900.0) # Groups: 8

Sol #	Pref ±	Cost ±	Time ±	Foods (top 5)	Nutrients vs DRIs
#1	-49.14	31.24	305	PORUMAL SUTU LD HACIK SUTU SAGSIKA MUSLAFMAN CORBASI FIZ	Energy(kcal): 2498 (2085.0-2400.0) # Protein(g): 58.78 (0-100.0) # Carb(g): 14 (175.0-399.0) # Fiber(g): 24 (0-9999.0) # SaturatedFat(g): 8.67 (1560.0-2900.0) # Groups: 10
#2	-136.76	26.73	470	BODURLER KECIL YEMER SUTU PASTIRMA PASTIRMA PASTIRMA	Energy(kcal): 2719 (2085.0-2400.0) # Protein(g): 58.78 (0-100.0) # Carb(g): 14 (175.0-399.0) # Fiber(g): 11 (0-9999.0) # SaturatedFat(g): 8.67 (1560.0-2900.0) # Groups: 10
#3	-178.36	31.23	305	AYNA MURU TIL SHARITIL HAVUC CORBASI YEMER SUTU PASTIRMA PASTIRMA	Energy(kcal): 2712 (2085.0-2400.0) # Protein(g): 58.78 (0-100.0) # Carb(g): 14 (175.0-399.0) # Fiber(g): 11 (0-9999.0) # SaturatedFat(g): 8.67 (1560.0-2900.0) # Groups: 10

Figure 3: Menu Tables

User Preferences & Dietary Profiles

- User 1 is considered as non-vegetarian.
- User 2 is considered as vegetarian.
- If explicit dietary information is unavailable, Preference and Preference2 columns in the Foods table are analyzed.
- Vegetarian users are detected by examining ratings for meat-based food items.
- Food items rated as -1 are strictly forbidden for the corresponding user.
- During fitness evaluation, any chromosome containing forbidden items receives a very large penalty value.

5. NUTRITIONAL CONSTRAINTS

The optimization process considers user-specific nutritional requirements based on Dietary Reference Intake (DRI) values.

The nutritional constraints ensure that generated meal plans satisfy minimum and maximum daily nutrient requirements.

The considered nutrients include:

- Calories
- Protein
- Carbohydrates
- Fat
- Fiber
- Sodium

Each user profile has different nutritional limits depending on dietary characteristics and restrictions.

For example:

- Vegetarian users cannot consume meat-based products.
- Non-vegetarian users have fewer dietary restrictions.

The system checks whether generated menus satisfy nutritional constraints during fitness evaluation.

#	Nutrient	Unit
C1	Energy	kcal
C2	Protein	g
C3	Carbohydrate	g
C4	Fiber (Fiber_total_dietary)	g
C5	Sodium (Na)	mg

Table 4: Nutritional Constraints

6. CHROMOSOME REPRESENTATION & DECODING

In evolutionary optimization, each candidate solution is represented as a chromosome.

The chromosome structure determines how meal plans are encoded and manipulated during the optimization process.

In this project, chromosomes are represented as permutations of food items.

The chromosome is divided into two sections:

- Breakfast genes

- Lunch/Dinner genes

The first 94 genes represent breakfast food items, while the remaining genes correspond to lunch and dinner foods.

The evolutionary algorithms use crossover and mutation operators to generate new meal combinations.

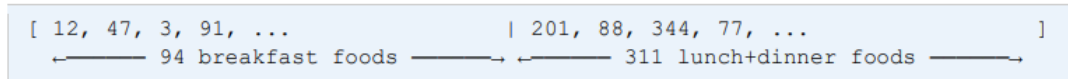


Table 5: Representation

7. DECODING (GENOTYPE → PHENOTYPE)

The decoding process converts chromosomes (genotype) into actual meal plans (phenotype).

During decoding:

1. Food items are selected according to chromosome order.
2. Nutritional values are accumulated.
3. Meal composition rules are checked.
4. Feasible menus are generated.

The decoding mechanism ensures that generated menus satisfy meal category requirements while maximizing nutritional feasibility.

The phenotype representation includes:

- Selected breakfast meals
- Selected lunch meals
- Selected dinner meals
- Total nutritional values
- Total cost
- Total preparation time

```

Upper bound: effective RUL = RUL × 1.15    (allow 15% over)
Lower bound: effective RLL = RLL × 0.90    (allow 10% under)

Breakfast split: RLL_b = RLL × 0.35    |    RUL_b = RUL × 0.35
  
```

Table 6: Mathematical Model

8. CONSTRAINTS & PENALTY FUNCTION

Constraint handling is one of the most important components of the optimization process.

If a candidate solution violates nutritional constraints, the system applies a penalty value to reduce the solution quality.

The penalty mechanism helps evolutionary algorithms avoid infeasible solutions.

The penalty is added to objective values such as:

- Cost
- Preparation time
- Preference score

The fitness function can be represented as:

Fitness = ObjectiveValue + Penalty

where the penalty increases when nutritional constraints are violated.

If a food item has preference score -1 for a user, a very large penalty is added and the solution becomes infeasible.

The system penalizes:

- Nutrient deficiency
- Excess nutrient intake
- Dietary restriction violations

Suggested Equation

Penalty Function:

$P(x) = \sum \text{Constraint Violations}$

```
For each nutrient j = 1..5, with menu total vj:  
  
viol_low_j = max(0, RLL_j - vj) / (RUL_j - RLL_j)  
viol_high_j = max(0, vj - RUL_j) / (RUL_j - RLL_j)  
  
R = 0.7 ×  $\sum$  viol_low_j + 0.3 ×  $\sum$  viol_high_j  
  
penalized_fitness = objective_value -  $\lambda$  × R  
( $\lambda$  = penalty weight, tune experimentally, start with  $\lambda$  = 1.0)
```

Table 7: Constraints & Penalty Function

9. DIVERSITY

Maintaining diversity is essential in multi-objective evolutionary optimization because it allows algorithms to explore different regions of the Pareto front.

A diverse solution set provides users with multiple alternative meal plans instead of a single solution.

In this project, diversity is maintained using evolutionary operators such as:

- Mutation
- Crossover
- Selection strategies

Algorithms like NSGA-II and NSGA-III explicitly use diversity preservation techniques to avoid premature convergence.

Diversity evaluation helps determine whether algorithms produce well-distributed Pareto-optimal solutions.

Option	Approach	Note
A — Extra objective	$f_diversity = \text{number of distinct foodGroupId values in menu or number of distinct ingredients in menu} \rightarrow \text{MAX}$	Makes problem 4-objective
B — Penalty term	$R_total = R + \alpha \times (1 / \text{distinct_group_count})$ or $R_total = R + \alpha \times (1 / \text{distinct_ingredients})$	Penalizes low diversity
C — Hard constraint	$\text{distinct foodGroupId count} \geq 4$ (enforced during decode)	Simplest to implement

Table 8: Diversity

10. ALGORITHMS

This project compares four state-of-the-art Multi-Objective Evolutionary Algorithms.

10.1 NSGA-II

NSGA-II (Non-dominated Sorting Genetic Algorithm II) is one of the most widely used multi-objective evolutionary algorithms.

Main characteristics:

- Fast non-dominated sorting
- Crowding distance calculation
- Elitist selection strategy

NSGA-II maintains solution diversity while converging toward the Pareto front.

Algorithm	Type	Key Mechanism
NSGA-II	Pareto-based	Non-dominated sorting + Crowding distance
SPEA2	Pareto + Archive	Strength fitness + k-NN density + Archive truncation
IBEA	Indicator-based	Hypervolume indicator (I_{ϵ}^{+})
SMS-EMOA	Indicator-based	Hypervolume contribution selection
NSGA-III	Reference-point	Structured reference points on hyperplane

Table 9: Algorithms Implement and compare at least 2 of the following MOEAs:

10.2 SPEA2

SPEA2 (Strength Pareto Evolutionary Algorithm 2) uses an external archive to preserve non-dominated solutions.

Main characteristics:

- Fitness assignment based on dominance strength
- Archive-based elitism
- Density estimation

SPEA2 generally provides strong convergence performance.

10.3 SMS-EMOA

SMS-EMOA (S-Metric Selection Evolutionary Multi-Objective Algorithm) uses hypervolume contribution for selection.

Main characteristics:

- Hypervolume-based optimization
- High-quality Pareto front approximation
- Strong convergence ability

This algorithm focuses on maximizing Pareto front coverage.

10.4 NSGA-III

NSGA-III is an extension of NSGA-II designed for many-objective optimization problems.

Main characteristics:

- Reference-point-based selection
- Improved diversity maintenance

- Better scalability for multiple objectives

NSGA-III performs well when objective dimensionality increases.

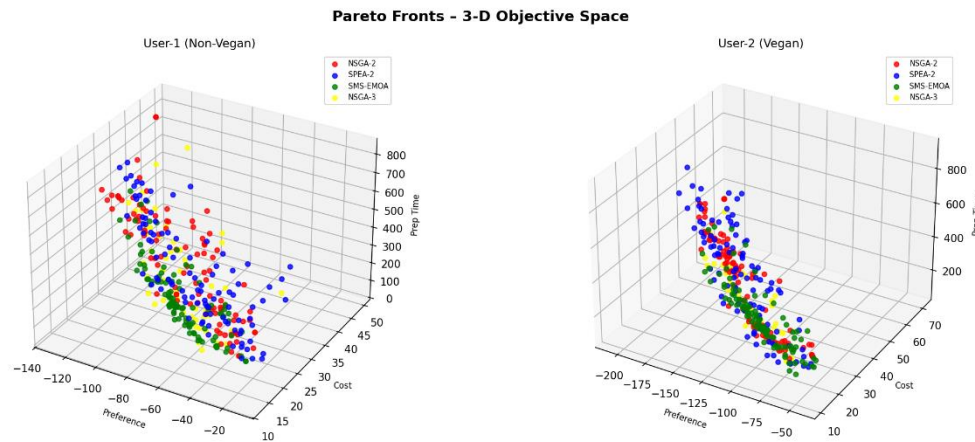


Table 10: Pareto Fronts

11. EXPECTED OUTPUTS & EVALUATION

The optimization system generates multiple Pareto-optimal daily menu recommendations.

Each solution represents a trade-off among:

- Nutritional adequacy
- User preference
- Cost
- Preparation time

The generated solutions are evaluated using:

- Hypervolume
- Convergence
- Diversity
- Constraint satisfaction

The algorithms are compared according to:

- Solution quality
- Feasibility
- Computational efficiency
- Pareto front distribution

#	Experiment	Description
1	User comparison	Run for both User 1 and User 2 — compare Pareto fronts
2	Algorithm comparison	Run ≥ 2 algorithms with same parameters — compare results
3	Diversity impact	Run with and without diversity mechanism — compare menu variety

Table 11: Required Experiments

Reference point: for each objective, take the worst value observed across all runs and add 10% margin.

All algorithms use the same reference point → results are comparable.

Report the reference point values used in your experiments.

Table 12: Hypervolume Calculation

Criterion	Metric
Constraint compliance	How many of 5 nutrients are within DRI bounds?
Preference quality	Total preference score of selected foods
Diversity	Number of distinct food groups in the menu
Objective values	Cost / time / CO ₂ depending on chosen objectives

Table 13: Menu Quality Metrics

13. RESULTS AND DISCUSSION

The experimental results demonstrate that evolutionary multi-objective optimization algorithms are effective for solving complex diet planning problems.

The generated Pareto-optimal solutions provide multiple meal plan alternatives with different trade-offs among objectives.

Among the evaluated algorithms:

- NSGA-II provides balanced diversity and convergence.
- SPEA2 achieves stable Pareto front approximations.
- SMS-EMOA performs strongly in hypervolume optimization.
- NSGA-III maintains diversity effectively for multiple objectives.

The vegetarian user profile introduces stricter constraints, making optimization more difficult compared to the non-vegetarian profile.

The comparison results highlight the importance of diversity preservation and constraint handling in diet optimization problems.

14. CONCLUSION

This study presented a multi-objective diet optimization system using evolutionary algorithms. The problem was modeled as a Multi-Objective Multidimensional Knapsack Problem (MOMKP), where personalized daily menus were generated under nutritional and dietary constraints.

Four evolutionary algorithms — NSGA-II, SPEA2, SMS-EMOA, and NSGA-III — were implemented and compared using two different user profiles.

The results showed that evolutionary multi-objective optimization techniques can successfully generate feasible and diverse meal plans while balancing nutritional adequacy, user preferences, cost, and preparation time.

Future work may include:

- Real-time user feedback integration
- Dynamic meal recommendation systems
- Larger food databases
- Deep learning-assisted preference prediction
- Mobile application integration